

Dear Illusie,

Let X be a complex analytic space, and let

$$\varepsilon: X_* \longrightarrow X$$

be a smooth simplicial analytic space which is a proper hypercovering of X .

Conjecture. The object

$$R\varepsilon_{**}(\Omega_{X_*}^p) \in D_{\text{coh}}^+(X, \mathcal{O})$$

is independent of the choice of X_* . Notation: $\underline{\Omega}_X^p$.

Remark. This greatly clarifies what I do in Hodge III. One has put X into a double complex, with d'_1 linear and d''_2 a differential operator of first order, and well defined up to filtered quasi-isomorphism; it is $R\varepsilon_{**}\Omega_{X_*}^*$, whose graded pieces are the $R\varepsilon_{**}\Omega_{X_*}^p$.

For X projective, the spectral sequence, degenerate at E_1 , which abuts to the Hodge filtration, is

$$H^q(X, \underline{\Omega}_X^p) \implies H^{p+q}(X^{\text{an}}, \mathbb{C}).$$

Evidence. By dévissage, it should no doubt suffice to see the following. Let X be smooth, let $\tilde{X} \xrightarrow{\pi} X$ be the blow-up of a smooth $Y \subset X$ of codimension c , and let $\tilde{Y} = \pi^{-1}(Y)$. Consider the system

$$(\tilde{X}/X)^{\Delta_n} \xrightarrow{\varepsilon_n} X,$$

where

$$(\tilde{X}/X)^k = \tilde{X} \text{ glued along } \tilde{Y} \text{ to } (\tilde{Y}/Y)^n$$

(the diagonal inclusion of $\tilde{Y}$). Let $\underline{\Omega}_{(\tilde{X}/X)^k}^p$ be defined as

$$\ker\left(\Omega_{\tilde{X}}^p \oplus \Omega_{(\tilde{Y}/Y)^n}^p \longrightarrow \Omega_{\tilde{Y}}^p\right).$$

Then

$$\Omega_X^p \xrightarrow{\sim} R\varepsilon_{**}\underline{\Omega}_{(\tilde{X}/X)^k}^p.$$

“Proof”. a) Let

$$q: (\tilde{Y}/Y)^{\Delta_k} \longrightarrow Y.$$

One has

$$Rq_{**}\Omega^p = \Omega^p$$

(clear: there are local sections).

b) This makes it possible to simplify the diagram. The geometric fact should be that X is the geometric realization of

$$\tilde{X} \rightrightarrows \tilde{X} \times_X \tilde{X},$$

and one must verify the triangle

$$0 \longrightarrow \Omega_X^p \longrightarrow \Omega_Y^p \oplus R\pi_*\Omega_{\tilde{X}}^p \longrightarrow R\pi'_*\Omega_{\tilde{Y}}^p \longrightarrow 0.$$

That is:

$$\Omega_X^p \simeq \pi_*\Omega_{\tilde{X}}^p, \quad \Omega_Y^p \simeq \pi'_*\Omega_{\tilde{Y}}^p,$$

and, for $i > 0$,

$$R^i \pi_* \Omega_{\tilde{X}}^p \simeq R^i \pi'_* \Omega_{\tilde{Y}}^p.$$

Indeed, one has

$$0 \longrightarrow \pi^* \Omega_X^1 \longrightarrow \Omega_{\tilde{X}}^1 \longrightarrow \Omega_{\tilde{Y}/Y}^1 \longrightarrow 0,$$

whence a filtration of $\Omega_{\tilde{X}}^p$ with successive quotients

$$\Omega_{\tilde{Y}/Y}^p, \quad \dots, \quad \Omega_{\tilde{Y}/Y}^p(i) \quad (0 < i < p), \quad \text{and} \quad \pi^* \Omega_X^p,$$

the former having $R\pi_* = 0$, and the latter having $R\pi_* = \Omega_X^p$. The assertion follows.

Example. For X projective, if $\underline{\Omega}_X^0 = \mathcal{O}$, then

$$H^*(X^{\text{an}}, \mathbf{C}) \longrightarrow H^*(X, \mathcal{O})$$

is surjective.

Example. $\mathcal{H}^0(\underline{\Omega}_X^0)$ is the subsheaf, on X , of the sheaf of rings of the normalization of X , formed by the functions that are constant on each reduced fiber. Denoting the normalization by n , one has

$$0 \longrightarrow \mathcal{H}^0(\underline{\Omega}_X^0) \longrightarrow \mathcal{O}_{X^n} \rightrightarrows \mathcal{O}_{(X^n \times_X X^n)_{\text{red}}}.$$

Example. One should have, for a finite group G acting on X , and $\pi: X \rightarrow X/G$,

$$\underline{\Omega}_{X/G} = (\pi_* \underline{\Omega}_X)^G.$$

Where $Z = \text{smooth/finite group}$, one should therefore have $\underline{\Omega}^0 = \mathcal{O}$. For a proper V-variety X , locally $X = X_1/G$, and

$$\underline{\Omega}_X^p = (\pi_* \Omega_{X_1}^p)^G,$$

the complex $\underline{\Omega}_X^\bullet$ then gives rise to a standard Hodge theory.

Example. If $A = B \cup_C D$, one should have a triangle

$$0 \longrightarrow \underline{\Omega}_A \longrightarrow \underline{\Omega}_B \oplus \underline{\Omega}_D \longrightarrow \underline{\Omega}_C \longrightarrow 0.$$

If $\underline{\Omega}^0 = \mathcal{O}$ for B , D , and C , it should therefore be the same for A .

Example. If $\tilde{X} \xrightarrow{\pi} X$ is birational, and an isomorphism outside $Y \subset X$, with inverse image $\tilde{Y}$, $\pi': \tilde{Y} \rightarrow Y$, one should have a triangle

$$0 \longrightarrow \underline{\Omega}_X \longrightarrow \underline{\Omega}_Y \oplus R\pi_* \underline{\Omega}_{\tilde{X}} \longrightarrow R\pi'_* \underline{\Omega}_{\tilde{Y}} \longrightarrow 0.$$

Example. Let X be the cone over a projective variety $V \subset \mathbf{P}^e$. One has:

$$\Gamma \mathcal{H}^0(\underline{\Omega}_X^0) = \mathbf{C} \oplus \bigoplus_{n>0} H^0(V, \mathcal{O}(n)),$$

and $\Gamma \mathcal{H}^1(\underline{\Omega}_X^0)$ is the following graded module over this graded algebra:

$$\bigoplus_{n>0} H^1(V, \mathcal{O}(n)).$$

One has $\underline{\Omega}_X^0 = \mathcal{O}$ if and only if, for $n > 0$,

$$H^*(\mathbf{P}^e, \mathcal{O}(n)) \longrightarrow H^*(V, \mathcal{O}(n))$$

is surjective. Example: $V = V_m(\underline{a})$, $\sum a_i < n + 2$ (FAC 78, Prop. 5).

Functoriality.

a) $\underline{\Omega}_X$ should be functorial in X .

b) There is a “cup product”, at least:

$$\mathcal{H}^i \underline{\Omega}^p \otimes \mathcal{H}^j \underline{\Omega}^q \longrightarrow \mathcal{H}^{i+j} \underline{\Omega}^{p+q}.$$

c) If X_* is a proper hypercovering of X , one should have

$$R\varepsilon_{**} \Omega_{X_*}^p = \underline{\Omega}_X^p.$$

Generalization.

a) Ideally, one should have such a theory for sheaves other than $\mathbf{C}$ (complexes of sheaves).

b) At least: $U \hookrightarrow X$, $Rj_* \mathbf{C}$ (cf. Hodge III).

I hope these scribbles are not too garbled.

Best regards,

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